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Methodology

Data and Documentation

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SECTION 1

Introduction

This document describes the methodology used to construct the variables available for download from the NEFIN website (<https://nefin.com.br>).

SECTION 2

Portfolios

We construct portfolios by sorting eligible BOVESPA stocks (see Section 3) by size, book-to-market ratio, momentum, illiquidity, and industry. We compute value-weighted and equal-weighted returns for the following portfolios:

- 3 portfolios sorted by size;
- 3 portfolios sorted by book-to-market;
- 3 portfolios sorted by momentum;
- 3 portfolios sorted by illiquidity;
- 4 portfolios sorted by size and book-to-market ratio (2-by-2);
- 4 portfolios sorted by momentum and size (2-by-2);
- 4 portfolios sorted by size and illiquidity (2-by-2);
- 7 portfolios sorted by industry;

Each portfolio group is available as an Excel workbook. The workbooks contain worksheets with value-weighted returns, equal-weighted returns, the number of stocks, average market value, average book value, and the average book-to-market ratio. The construction of these variables is described below.

2.1 Value-Weighted Returns (daily)

The value-weighted return of portfolio P on day t is computed as

$$R_t = \sum_i \omega_{i,t} r_{i,t}$$

where:

- $\omega_{i,t}$ is the weight of stock i on day t , defined as the ratio of the market value of stock i at $t - 1$ to the total market value of portfolio P at $t - 1$;
- $r_{i,t}$ is the return of stock i on day t , computed as

$$r_{i,t} = \begin{cases} 0, & \text{if stock } i \text{ is not traded on day } t, \\ \frac{P_t^i}{P_{t-1}^i} - 1, & \text{otherwise;} \end{cases}$$

- P_t^i is the price of stock i on day t , adjusted for dividends and stock splits.

2.2 Equal-weighted Returns (daily)

The equal-weighted return of portfolio P on day t is computed as

$$R_t = \frac{1}{N} \sum_i r_{i,t}$$

where:

- N is the number of stocks in the portfolio;
- $r_{i,t}$ is the return of stock i on day t , as defined in Section 2.1.

2.3 Number of Stocks (monthly)

The number of stocks in portfolio P in month t is the total number of stocks assigned to that portfolio.

2.4 Average Market Value (monthly)

The average market value of a portfolio in month t is the arithmetic mean of the market values of its constituent stocks, expressed in thousands of Brazilian reais. Each stock is assigned the market value of its issuing firm.

2.5 Average Book Value (annual)

The average book value of a portfolio in year t is the arithmetic mean of the book values of its constituent stocks, expressed in thousands of Brazilian reais. Each stock is assigned the book value of its issuing firm. We use book values measured in June.

2.6 Average Book-to-market (annual)

The average book-to-market ratio of a portfolio in year t is the arithmetic mean of the book-to-market ratios of its constituent stocks. Each stock is assigned the ratio of its issuing firm. We use ratios measured in June.

SECTION 3

Eligibility Criteria

A stock traded on BOVESPA is eligible in year t if it satisfies all three of the following criteria:

- It is the firm's most actively traded share class, measured by total trading volume during the previous year;
- It was traded on more than 80% of the trading days in year $t - 1$, with daily trading volume above R\$ 500,000. If the stock was listed during year $t - 1$, the measurement window runs from its listing date through the final trading day of that year;
- Its initial listing occurred before December of year $t - 1$.

SECTION 4

Portfolio Methodology

4.1 3 Portfolios Sorted by Size

Each January of year t , we sort eligible stocks in ascending order into terciles based on their December market capitalization in year $t - 1$. Each stock is assigned the market capitalization of its issuing firm. We hold the resulting portfolios throughout year t .

4.2 3 Portfolios Sorted by Book-to-Market

Each January of year t , we sort eligible stocks in ascending order into terciles based on their book-to-market ratios in June of year $t - 1$. Each stock is assigned the ratio of its issuing firm. We hold the resulting portfolios throughout year t .

4.3 3 Portfolios Sorted by Momentum

Each month t , we sort eligible stocks in ascending order into terciles based on their cumulative returns from month $t - 12$ through month $t - 2$. We hold the resulting portfolios during month t .

4.4 3 Portfolios Sorted by Illiquidity

Each month t , we sort eligible stocks in ascending order into terciles based on their trailing 12-month average illiquidity. Stock illiquidity is computed following Acharya and Pedersen (2005). We hold the resulting portfolios during month t .

4.5 4 Portfolios Sorted by Size and Book-to-Market (2-by-2)

Each January of year t , we independently sort eligible stocks in ascending order by size and book-to-market ratio, following Sections 4.1 and 4.2. We hold the resulting four portfolios throughout year t .

4.6 4 Portfolios Sorted by Size and Momentum (2-by-2)

Each month t , we independently sort eligible stocks in ascending order by size and momentum, following Sections 4.1 and 4.3; the size sort is updated monthly. We hold the resulting four portfolios during month t .

4.7 4 Portfolios Sorted by Size and Illiquidity (2-by-2)

Each month t , we independently sort eligible stocks in ascending order by size and illiquidity, following Sections 4.1 and 4.4; the size sort is updated monthly. We hold the resulting four portfolios during month t .

4.8 7 Portfolios Sorted by Industry

We classify eligible stocks into the following industry sectors: Basic Products, Construction, Consumer, Energy, Finance, Manufacturing, and Other.

SECTION 5

Risk Factors

5.1 Market Factor

The market factor is the difference between the value-weighted daily return on the market portfolio, which includes all eligible stocks, and the daily risk-free rate. We derive the daily risk-free rate from the 30-day DI swap rate.

5.2 Small Minus Big (SMB)

The *Small Minus Big (SMB)* factor is the return on a portfolio that is long stocks with low market capitalization (“Small”) and short stocks with high market capitalization (“Big”).

Each January of year t , we sort eligible stocks in ascending order into three groups based on their December market capitalization in year $t - 1$. We then compute the equal-weighted returns on the first (“Small”) and third (“Big”) portfolios. SMB is the return on the Small portfolio minus the return on the Big portfolio.

5.3 High Minus Low (HML)

The *High Minus Low (HML)* factor is the return on a portfolio that is long stocks with high book-to-market ratios (“High”) and short stocks with low book-to-market ratios (“Low”).

Each January of year t , we sort eligible stocks in ascending order into three groups based on their firms’ book-to-market ratios in June of year $t - 1$. We then compute the equal-weighted returns on the first (“Low”) and third (“High”) portfolios. HML is the return on the High portfolio minus the return on the Low portfolio.

5.4 Winners Minus Losers (WML)

The *Winners Minus Losers (WML)* factor is the return on a portfolio that is long stocks with high past returns (“Winners”) and short stocks with low past returns (“Losers”).

Each month t , we sort eligible stocks in ascending order into three groups based on their cumulative returns from month $t - 12$ through month $t - 2$. We then compute the equal-weighted returns on the first (“Losers”) and third (“Winners”) portfolios. WML is the return on the Winners portfolio minus the return on the Losers portfolio.

5.5 Illiquid Minus Liquid (IML)

The *Illiquid Minus Liquid (IML)* factor is the return on a portfolio that is long stocks with high illiquidity (“Illiquid”) and short stocks with low illiquidity (“Liquid”).

Each month t , we sort eligible stocks in ascending order into three groups based on their trailing 12-month average illiquidity, computed following Acharya and Pedersen (2005). We then compute the equal-weighted returns on the first (“Liquid”) and third (“Illiquid”) portfolios. IML is the return on the Illiquid portfolio minus the return on the Liquid portfolio.

SECTION 6

Illiquidity Index

The illiquidity of stock i measures how strongly its price moves in response to trading volume. We construct the measure following Acharya and Pedersen (2005):

$$ILLIQ_t^i = \min \left\{ \frac{1}{Days_t^i} \sum_{d=1}^{Days_t^i} \frac{|r_{td}^i|}{V_{td}^i / P_{t-1}^M}, 30.00 \right\},$$

where:

- $Days_t^i$ is the number of days on which stock i was traded in month t ;
- r_{td}^i is the return of stock i on day d of month t , as defined in Section 2.1;
- V_{td}^i is the trading volume, expressed in millions, of stock i on day d of month t ;
- P_{t-1}^M is the ratio of the market portfolio's capitalization at the end of month $t - 1$ to its capitalization at the end of January 2000.

Capping rule

The stock-level illiquidity measure is capped at 30.00 before aggregation, as shown in the formula above. This cap limits the influence of extreme stock-level observations.

The Illiquidity Index is the value-weighted average illiquidity of all eligible stocks and therefore represents aggregate market illiquidity in Brazil.

SECTION 7

Cost of Equity

We estimate the cost of equity for each sector represented in the seven industry portfolios using the CAPM-based procedure described below.

We use NEFIN data to regress each industry's monthly excess return on the monthly market factor. For projects with horizons of 1, 5, 10, and 20 years, we multiply the estimated betas by the corresponding monthly U.S. market risk premium and add the relevant real risk-free rate.

7.1 US market risk premium

For each maturity, we calculate the risk premium by compounding annual U.S. market excess returns over a rolling window of the corresponding length.

7.2 Real risk-free rate

We derive the one-year real risk-free rate from the 360-day DI swap rate, deflated by expected inflation measured by the IPCA index. The underlying data are available from the Central Bank of Brazil.

For the remaining maturities, we use spline interpolation of NTN-B yields.

SECTION 8

Volatility Index

We compute IVol-Br, a daily volatility index for the Brazilian market based on Carr and Wu (2006).

IVol-Br measures the expected volatility of the BOVESPA index (IBOVESPA) over two months, or 42 business days.¹ It is the weighted average of the near-term and next-term volatilities implied by options on the IBOVESPA spot index. On date t , the near term is the closest available option expiration, and the next term is the immediately subsequent expiration.²

Near-term and next-term variances are calculated as follows:

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{j}{T} \left[\frac{F}{K_0} - 1 \right]^2 \quad (1)$$

where:

- σ = $\frac{\text{volatility}}{100}$;
- T : time until expiration;
- F : forward level of the IBOVESPA, equal to its daily settlement price;
- K_0 : the strike closest to the forward level F ;
- K_i : strike of the i -th out-of-the-money option—a call when $K_i > K_0$, a put when $K_i < K_0$, and both when $K_i = K_0$;
- ΔK_i : the strike interval, equal to half the difference between the strikes immediately above and below K_i :

$$\Delta K_i = \frac{K_{i+1} - K_{i-1}}{2}$$

- R : the risk-free interest rate through expiration T , derived from the daily settlement price of the interbank deposit (DI) futures rate;
- $Q(K_i)$: the market price of the option with strike K_i ;
- j : an adjustment factor ($j = 0, 1$, or 2), determined by the following rule:

	$K_0 < F$	$K_0 > F$
$\exists$ call, $\exists$ put	$j = 1$	$j = 1$
$\exists$ call, $\nexists$ put	$j = 2$	$j = 0$
$\nexists$ call, $\exists$ put	$j = 0$	$j = 2$

This adjustment converts an in-the-money call (put) into the corresponding out-of-the-money put (call).

After calculating the near-term and next-term variances, we aggregate them into the IVol-Br measure published on the NEFIN website:

¹The index in Carr and Wu (2006) measures one-month expected volatility. For Brazil, the measure is restricted to a two-month horizon because IBOVESPA options expire only in even-numbered months.

²For example, on a date in January 2015, the near-term expiration was in February 2015 and the next-term expiration was in April 2015.

$$\text{IVol-Br} = 100 \times \sqrt{\left\{ T_1 \sigma_1^2 \left[\frac{N_{T_2} - N_{42}}{N_{T_2} - N_{T_1}} \right] + T_2 \sigma_2^2 \left[\frac{N_{42} - N_{T_1}}{N_{T_2} - N_{T_1}} \right] \right\} \times \frac{N_{252}}{N_{42}}} \quad (2)$$

- N_{T_1} : number of minutes until the near-term expiration;
- N_{T_2} : number of minutes until the next-term expiration;
- N_{42} : number of minutes in 42 business days (42×1440);
- N_{252} : number of minutes in 1 business year (252×1440).

Extrapolation rule

Equation (2) becomes an extrapolation when the weight assigned to the next term is negative. This situation occurs immediately after the near-term options expire, when the new near-term expiration is more than 42 business days away.

We make the following additional adjustments to accommodate features of the Brazilian options market:

- We restrict the options sample to trades executed between 3:00 p.m. and 6:00 p.m.;
- For each ticker, we retain only the final trade within that interval;
- We calculate near-term volatility only when the sample contains at least two call trades at different strikes and two put trades at different strikes. The same requirement applies to next-term volatility. This rule reduces errors caused by limited options-market liquidity;
- If near-term volatility is unavailable on a given day, IVol-Br equals next-term volatility. Conversely, if next-term volatility is unavailable, IVol-Br equals near-term volatility. If neither measure is available, IVol-Br is reported as missing;
- When the weight on the second term of Equation (2) is negative, we disregard next-term volatility and set IVol-Br equal to near-term volatility.

SECTION 9

Predictability

9.1 Dividend Yield

A stock's dividend yield is the ratio of dividends paid over a specified period to the stock price. We calculate an aggregate dividend yield for the Brazilian market using only eligible stocks. In each month m , we sum dividends and interest on shareholders' equity paid during the preceding 12 months, excluding month m itself (months $m - 12$ through $m - 1$). The dividend yield on day t is the ratio of this aggregate payment to the total market value of equity on that day.

SECTION 10

Stock Loans

All weighted averages in this section use the value of open positions as the common weighting variable. Let $OPV_{i,t}$ denote the value of open stock-loan positions in security i at time t . For the n_t eligible securities included in an aggregate measure, the normalized weight is

$$\omega_{i,t}^{\text{OPV}} = \frac{OPV_{i,t}}{\sum_{j=1}^{n_t} OPV_{j,t}}. \quad (3)$$

10.1 Short Interest

A stock's short interest is the ratio of shares sold short to shares outstanding. We measure shares sold short using shares on loan:

$$SI_{i,t} = \frac{Q_{i,t}}{O_{i,t}}, \quad (4)$$

where $Q_{i,t}$ is the number of shares of stock i on loan at time t and $O_{i,t}$ is the corresponding number of shares outstanding.

We calculate aggregate short interest for the Brazilian market as an average across eligible stocks, weighted by open-position value:

$$SI_t^{\text{agg}} = \sum_{i=1}^{n_t} \omega_{i,t}^{\text{OPV}} SI_{i,t}. \quad (5)$$

10.2 Days to Cover

Days to Cover measures the number of trading days theoretically required to close all outstanding short positions, based on the stock's average daily trading volume. For each stock, it is defined as

$$DTC_{i,t} = \frac{Q_{i,t}}{\bar{V}_{i,t}^{30}}, \quad (6)$$

where $Q_{i,t}$ is the number of shares on loan, as defined above, and $\bar{V}_{i,t}^{30}$ is the average daily trading volume of stock i , measured in shares, over the previous 30 trading days.

We calculate aggregate Days to Cover as an average across eligible stocks, weighted by open-position value:

$$DTC_t^{\text{agg}} = \sum_{i=1}^{n_t} \omega_{i,t}^{\text{OPV}} DTC_{i,t}. \quad (7)$$

10.3 Loan-Fee

The loan fee is the weekly average securities lending fee for Brazilian equities traded in the BTC market, expressed as an annualized fraction. Let $LF_{i,t}$ denote the loan fee for security i at time t . The aggregate loan fee index is the average weighted by open-position value:

$$LF_t^{\text{OPV}} = \sum_{i=1}^{n_t} \omega_{i,t}^{\text{OPV}} LF_{i,t}. \quad (8)$$

Securities with a larger value of open positions therefore have greater influence on the aggregate loan fee index.

SECTION 11

References

Acharya, V. V. and L. H. Pedersen (2005). "Asset Pricing with Liquidity Risk". In: *Journal of Financial Economics* 77.2, pp. 375-410.

Carr, P. and L. Wu (2006). "A Tale of Two Indices". In: *Journal of Derivatives*, pp. 13-29.

Contact

Questions or comments should be sent to nefin@usp.br.